

Shared Augmented Reality Experience Between a Microsoft Flight Simulator User and a User in the Real World

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VIDEOLINK

<https://youtu.be/ngPJNtdsviU>

ABSTRACT

Our video shows an application where a user with an augmented reality (AR) display watches another user, flying an airplane in the Microsoft Flight Simulator 2020 (MSFS), at their respective location in the real world. To do that, we take the location data of a virtual 3D airplane model in a virtual representation of the world of a user playing MSFS, and stream it via a server to a mobile AR device. The mobile device user can then see the same 3D airplane model at exactly that real world location, that corresponds to the location of the virtual 3D airplane model in the virtual representation of the world. The mobile device user can also see the plane movement updated according to the 3D airplane movement in the virtual world. We implemented the application on both a cellphone and the Hololens 2.

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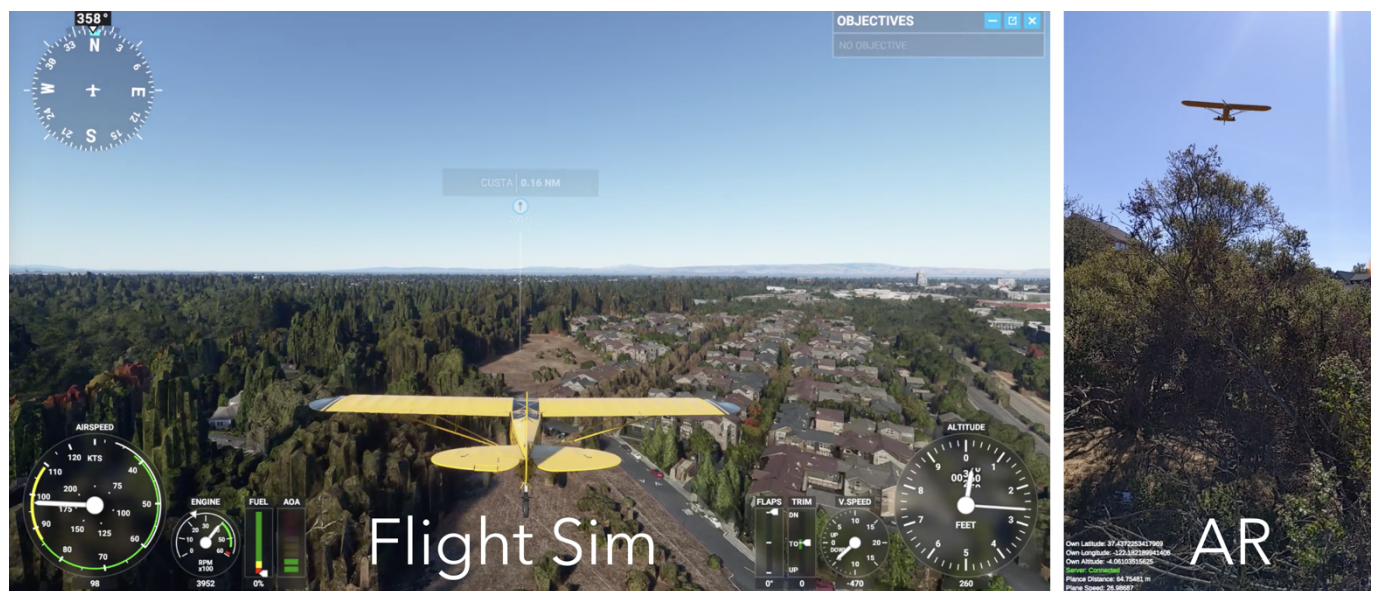


Figure 1: (Left) A view of a 3D model of a virtual airplane flying in Microsoft Flight Simulator 2020. (Right) A view of the app showing the camera stream of the real world and, in the sky, a 3D rendering of the same airplane as in the flight simulator flying in the sky. The location of the 3D model of the airplane in the app is the same world position as the position of the 3D model in the virtual world of the flight simulator.